

5. Second order differential equations

What students need to learn:

The linear second order differential equation $a\frac{d^2y}{dx^2} + b\frac{dy}{dx} + cy = f(x)$

where a , b and c are real constants and the particular integral can be found by inspection or trial.

The auxiliary equation may have real distinct, equal or complex roots. $f(x)$ will have one of the forms ke^{px} , $A + Bx$, $p + qx + cx^2$ or $m \cos \omega x + n \sin \omega x$.

Students should be familiar with the terms 'complementary function' and 'particular integral'.

Students should be able to solve equations of the form

$$\frac{d^2y}{dx^2} + 4y = \sin 2x.$$

Differential equations reducible to the above types by means of a given substitution.
